Data Warehouse Integration

SoftUni Team

Technical Trainers

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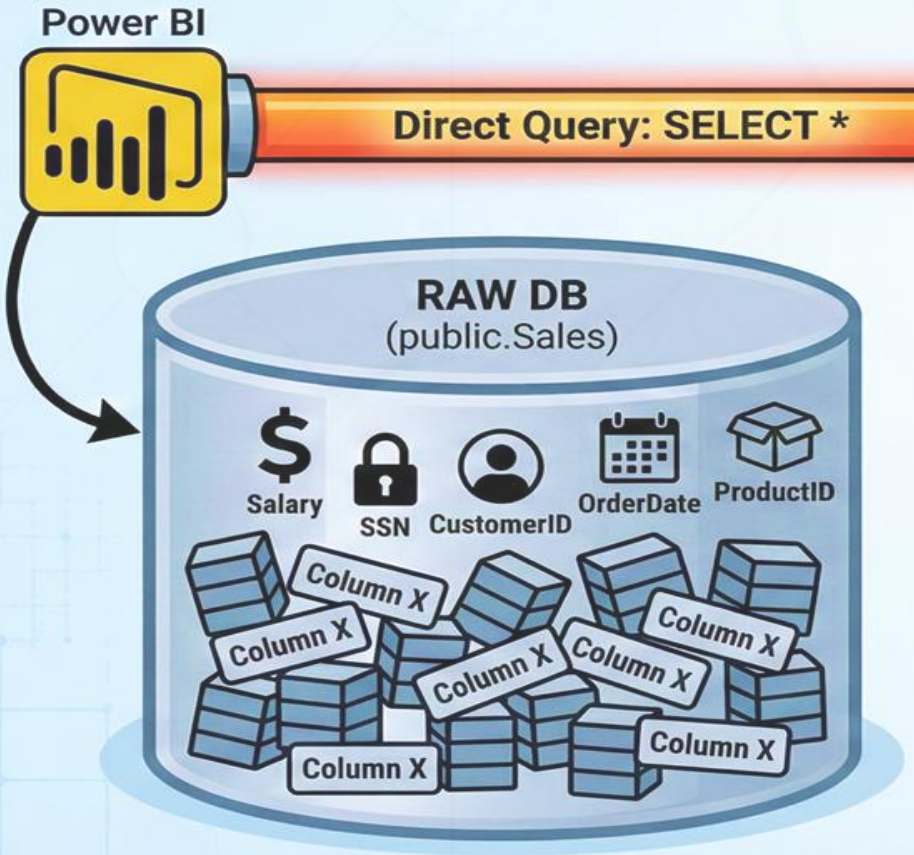
THE ROOKIE MISTAKE: Connecting to Raw Tables

The Rookie Mistake

Scenario: You connect Power BI directly to the Sales table. "SELECT * FROM **public.Sales**"



BREAKING CHANGES

If a DBA renames a column, report **breaks**.

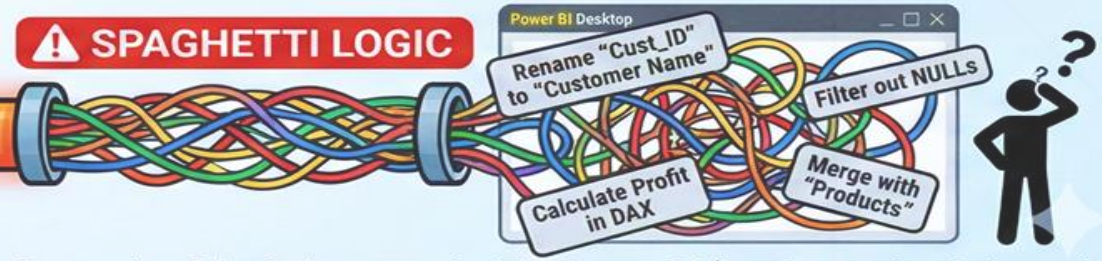
PERFORMANCE

You load **unnecessary columns**. Slows everything down.

SECURITY

Exposes sensitive data (e.g., Salary, SSN) to the report.

SPAGHETTI LOGIC



Renaming & logic happens inside Power BI (hard to maintain/reuse).

The Rookie Mistake

- **Scenario:** You connect Power BI directly to the raw table public.sales_data.
- **The Query:** SELECT * FROM sales_data.
- **Result:** You import 50 columns, even if you need only 3.

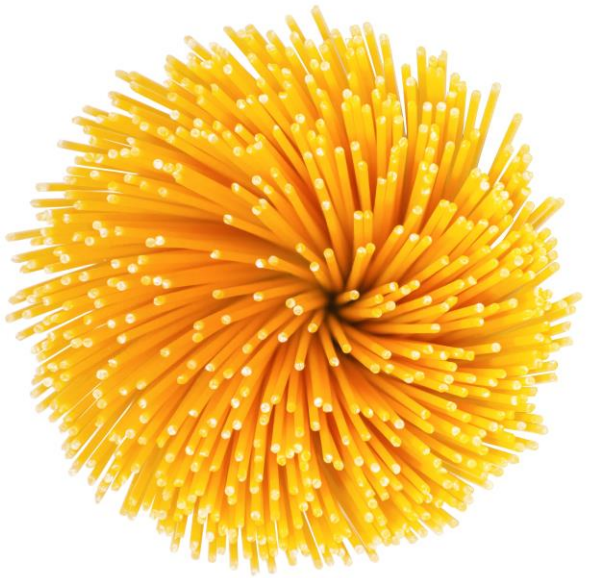
Why is SELECT * Dangerous?

- **Performance:** Wastes memory and network bandwidth.
- **Security:** Might expose salary, personal_id, or profit_margin to unauthorized users.
- **Fragility:** If the database admin renames

a column, your report crashes.

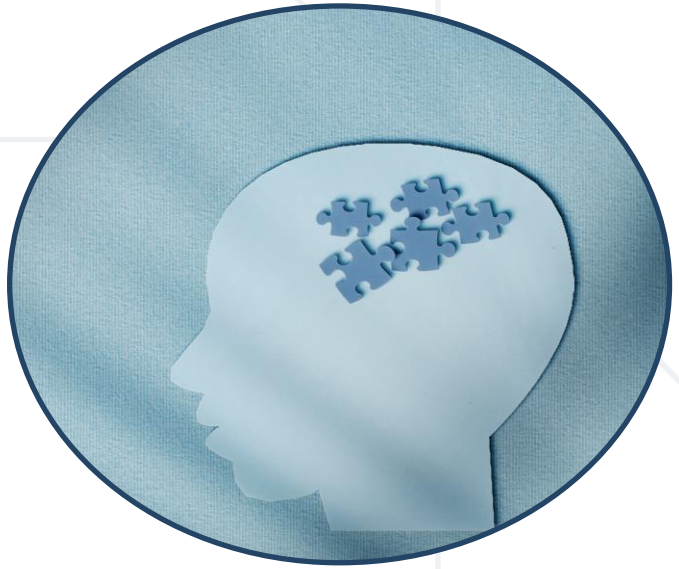
The Logic Trap

- **Scenario:** You rename "cust_nm" to "Client Name" inside Power BI.
- **Problem:** You have to do this in every single report.
- **Result:** Inconsistent logic (Spaghetti code).



The Solution: Upstream Thinking

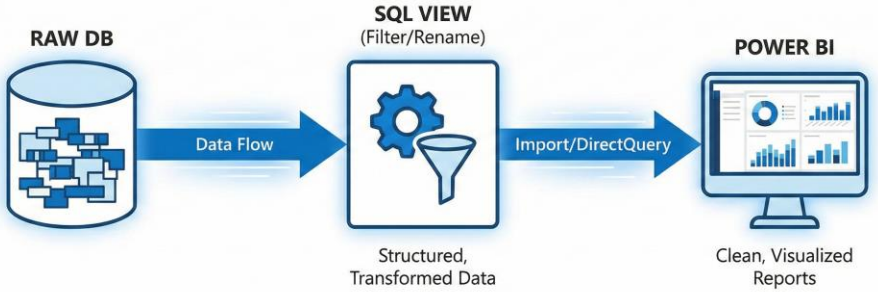
- **Downstream (Bad):** Fixing data in Power BI.
- **Upstream (Good):** Fixing data in the Database (SQL).
- **Rule:** Push the logic as close to the source as possible.



The Abstraction Layer

Definition: A shield between the messy database and the clean report.

Implementation: We use SQL Views.



Live Demo

What is a View?

- **Definition:** A "Virtual Table".
- It takes 0 bytes of storage space.
- It is simply a saved query that runs every time you request it.

Benefits of Views

1. **Security:** Expose only specific columns.
2. **Simplicity:** Hide complex JOINs from the end user.
3. **Consistency:** Calculate Total Revenue once in SQL, use it everywhere.

Syntax: Creating a View

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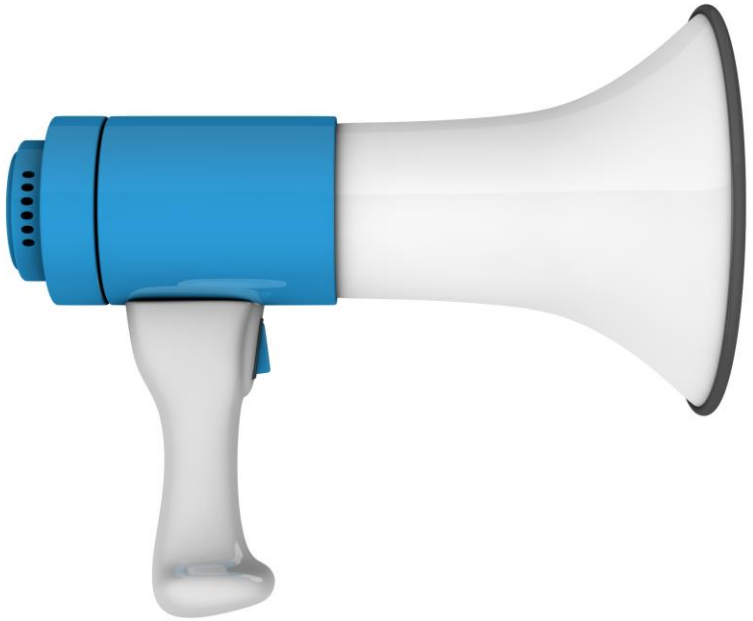
Standard SQL:

```sql
CREATE VIEW view_name AS
SELECT column1, column2
FROM table_name;
```

Best Practice:

Always use **OR REPLACE** to easily update it later.

PostgreSQL Specifics - Case Sensitivity



Warning: PostgreSQL converts everything to lowercase unless you use quotes.

`Select Name From Users` -> looks for `name` in `users`.

`Select "Name" From "Users"` -> looks for `Name` in `Users`.

Handling Quotes in Views

- If your columns have spaces or CamelCase,
- you MUST use double quotes:

```sql
SELECT
    first_name AS "First Name",  -- Clean name for Power BI
    last_name AS "Last Name"
FROM employees
```

```
-- ❌ Without quotes (PostgreSQL converts to lowercase)
SELECT Name FROM Users  → searches for: name, users

-- ✅ With quotes (preserves case)
SELECT "Name" FROM "Users"  → searches for: Name, Users
```

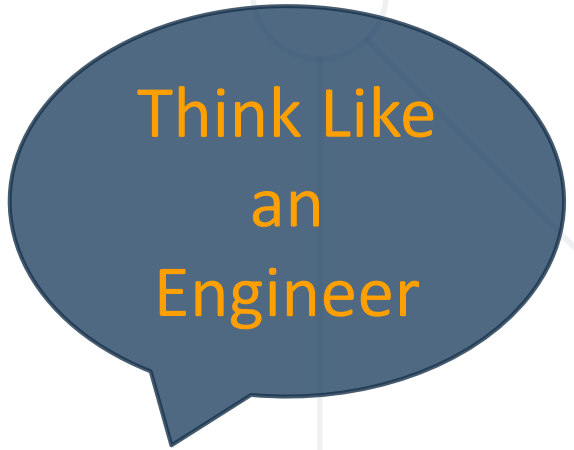
Naming Conventions

- Use prefixes to organize your views.
- vw_dim_ -> Dimension (Customers, Products, Locations).
- vw_fct_ -> Fact (Sales, Orders, Transactions).
- **Example:** vw_dim_Customers, vw_fct_Orders.

Naming Conventions

Your view names should tell a story.

- **vw_dim_Customer**
  - *vw* = View
  - *dim* = Dimension (Descriptive data: Who, Where)
- **vw_fct_Sales**
  - *fct* = Fact (Transactional data: Numbers, Dates)
- **Golden Rule:** Column names in the View should be user-friendly.
- Avoid: t_stamp_cr
- Use: Creation Date



Transformation in SQL

- Don't clean in Power BI if you can clean in SQL.
- Don't just select. Transform!
- Concatenation: `first_name || ' ' || last_name`
- Math: `qty * price`
- Filtering: `WHERE is_active = true`

Transformation in SQL

Bad Practice 👎

(Doing it in Power Query):

- Renaming cust_fname -> First Name
- Calculating Price * Qty

❌ Power Query M code (messy)
✅ SQL View (clean)

Good Practice 👍

(Doing it in the View)

```sql
CREATE VIEW vw_Order_Details AS
SELECT
    o.order_id,
    c.customer_name AS "Client",  -- Renaming
    o.qty * o.price AS "Total Revenue" -- Calculation
FROM orders o
JOIN customers c ON o.cust_id = c.id
```

Live Demo

Entering the Power BI World

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What is Power Query?

- It is the **Kitchen** of Power BI.
- It extracts, cleans, and prepares data before it goes to the dashboard.
- *Note: It is the same engine used in Excel.*

- **Tool:** Power BI Desktop.
- **Engine:** Power Query.

The Interface

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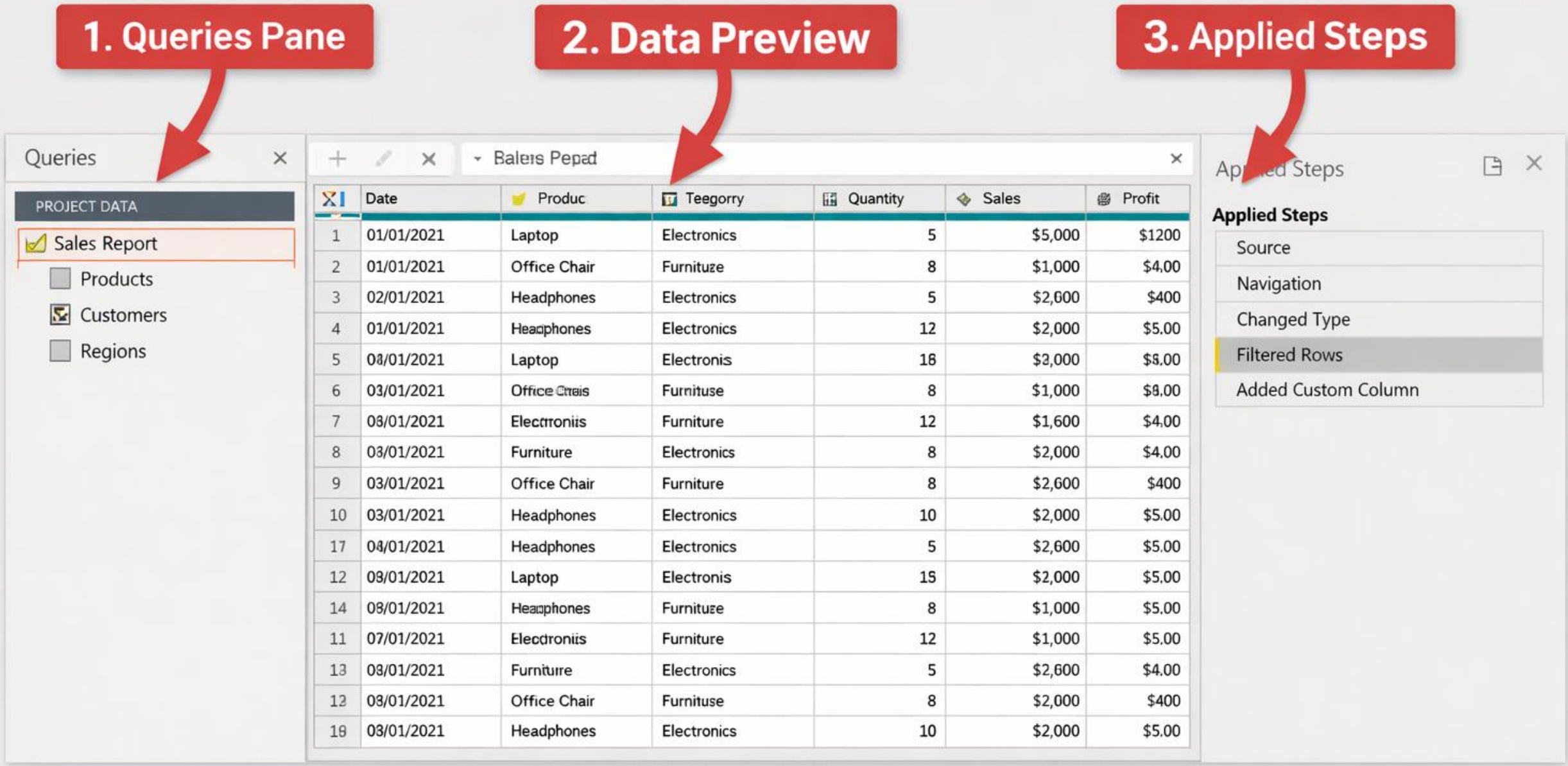
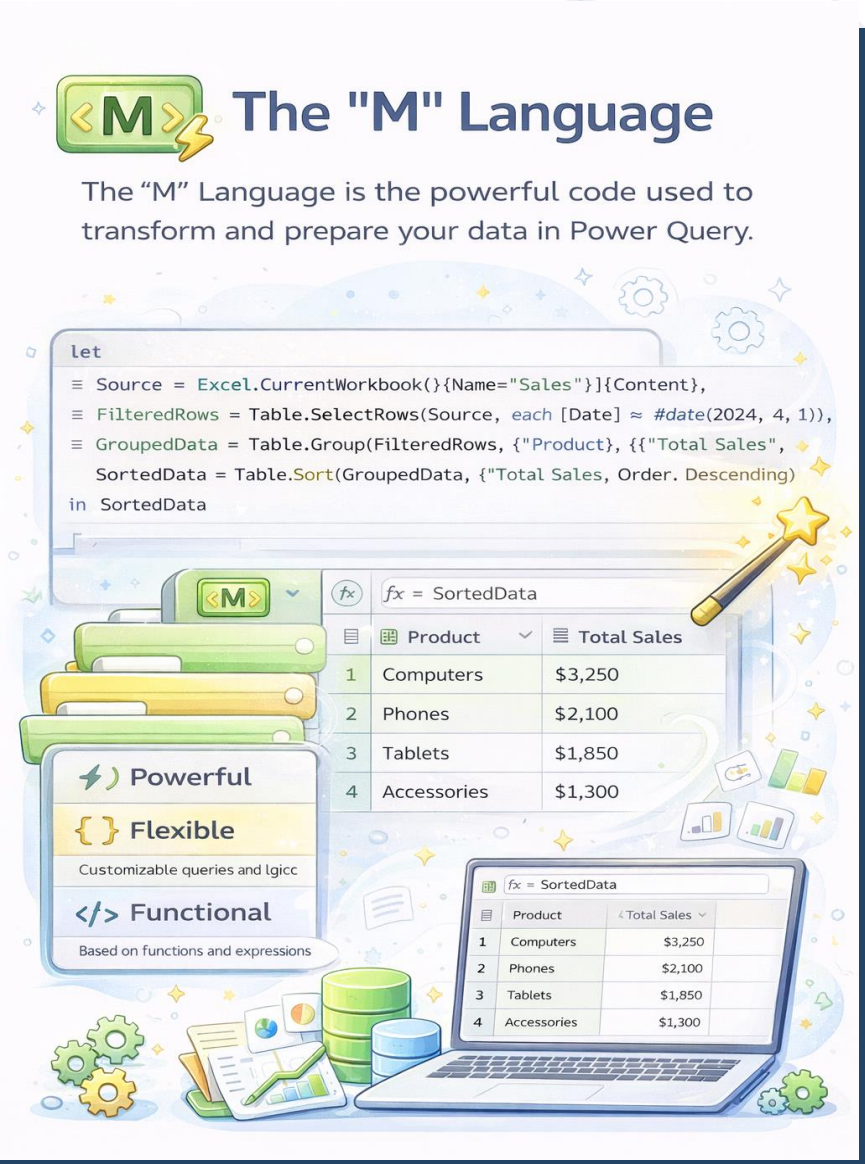


	Date	Produc	Teegorry	Quantity	Sales	Profit
1	01/01/2021	Laptop	Electronics	5	$5,000	$1200
2	01/01/2021	Office Chair	Furnituze	8	$1,000	$4,00
3	02/01/2021	Headphones	Electronics	5	$2,600	$400
4	01/01/2021	Headphones	Electronics	12	$2,000	$5.00
5	08/01/2021	Laptop	Electronis	18	$3,000	$8.00
6	03/01/2021	Office Chais	Furnituse	8	$1,000	$8.00
7	08/01/2021	Electroniis	Furniture	12	$1,600	$4,00
8	03/01/2021	Furniture	Electronics	8	$2,000	$4.00
9	03/01/2021	Office Chair	Furniture	8	$2,600	$400
10	03/01/2021	Headphones	Electronics	10	$2,000	$5.00
17	04/01/2021	Headphones	Electronics	5	$2,600	$5.00
12	09/01/2021	Laptop	Electronis	15	$2,000	$5.00
14	08/01/2021	Headphones	Furnituze	8	$1,000	$5.00
11	07/01/2021	Electroniis	Furniture	12	$1,000	$5.00
13	08/01/2021	Furniture	Electronics	5	$2,600	$4.00
12	03/01/2021	Office Chair	Furnituse	8	$2,000	$400
18	03/01/2021	Headphones	Electronics	10	$2,000	$5.00

The "M" Language



- Every button you click generates code.
- This code is called M (Mashup Language).
- You don't need to write it fluently, but you need to read it.

Applied Steps = Time Travel

- You can click on any previous step to see how data looked then.
- You can insert, delete, or reorder steps.
- Warning: Deleting an early step might break later steps.

Connecting to PostgreSQL

- Get Data -> Database -> PostgreSQL.
- Server: localhost (or IP).
- Database: TechCorp_DB.
- Data Connectivity: Import (usually).

The Golden Rule

- "Filter Early, Filter Often.“
- Remove data you don't need as soon as possible.

Live Demo

What is Query Folding?

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Query Folding: The Performance Magic

Definition:

- The ability of Power Query to generate a single SQL statement that executes on the server side.
- Power Query translates your UI clicks into a native SQL Statement.
- It sends this SQL to the database.
- The Database does the work, not your laptop.

Translation:

- **You do:** Filter rows in Power BI (Keep last 2 years).
- **Power BI does:** Sends a WHERE Year > 2023 to the database.
- **Result:** Only the relevant data travels over the network.

Analogy: The Pizza Order

Folded (Good):

- Simple filters, renaming, basic math.
- Power BI writes SQL for you.

"I want a Pizza without onions.“

```
✅ Folded (Server-side):
   🍕 Chef removes onions → You get clean pizza
   💾 DB filters data → Power BI gets clean data
```

Not Folded (Bad):

- Writing complex custom SQL in the "Get Data" window.
- Using specific M functions (e.g., Index Column, some text parsing).
- Power BI downloads EVERYTHING and filters locally.

"I want a standard Pizza .“

```
❌ Not Folded (Client-side):
   🍕 You manually remove onions → Wasted time
   💾 Power BI filters locally → Wasted bandwidth
```

How to verify Folding?

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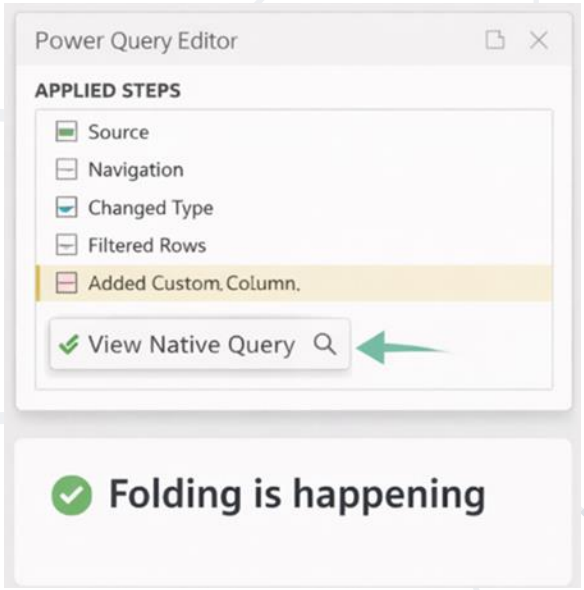
Validating Query Folding

Steps:

1. Go to **Power Query Editor**.
2. Right-click the last step in "Applied Steps".
3. Look for **"View Native Query"**.

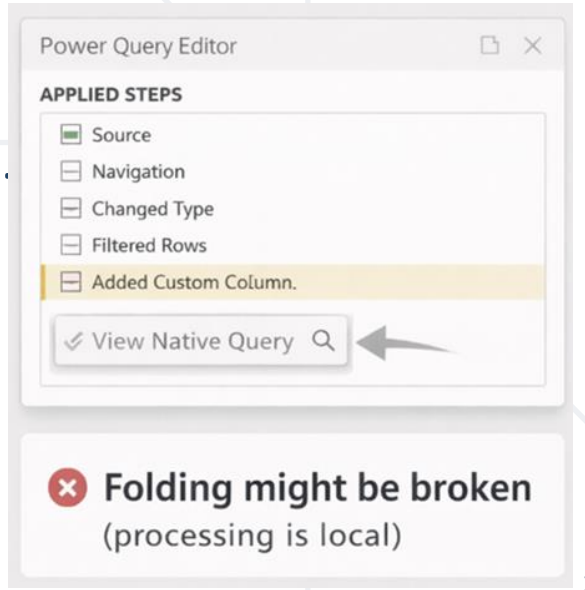
If Clickable:

Folding is happening.



If Greyed Out:

Folding might be broken (processing is local).



What breaks Folding?

- Writing Custom SQL in the source settings.
- Using transforms that SQL doesn't support (e.g., complex text parsing, index columns).
- Mixing data sources (Excel + SQL).

The "Custom SQL" Trap

If you paste a SQL query in the "Get Data" window.....

Power BI treats it as a "Black Box".

It stops folding immediately.

Solution: Create a View in the DB instead!

```
❌ Don't do this:
[Get Data → PostgreSQL → "Advanced Options"]
SELECT * FROM sales WHERE year = 2024

Result: No query folding! ❌

✅ Do this instead:
1. CREATE VIEW vw_sales_2024 AS SELECT * FROM sales WHERE year = 2024
2. Power BI → Connect to vw_sales_2024

Result: Query folding works! ✅
```

Live Demo

Optimization Checklist

Performance Checklist:

- ✅ Heavy logic in SQL Views
- ✅ Simple filters in Power Query (they fold)
- ✅ Check "View Native Query" regularly
- ❌ Avoid Custom SQL in source
- ❌ Avoid Index Column transforms
- ❌ Don't mix data sources (Excel + SQL)

Why care?

- **Folding:** 5 seconds refresh.
- **No Folding:** 5 minutes refresh + Laptop fan spinning + Memory crash.

1. Do heavy logic in SQL Views.
2. Do simple filters/renaming in Power Query (it folds).
3. Check "Native Query" often.

The Environment Problem

- You build a report connected to DEV_DB (Test data).
- Now you need to publish it for the CEO using PROD_DB (Real data).
- Do you manually edit 50 queries?

Parameters

- **Definition:**
  Variables that store configuration values.
- **Use them for:**
  Server Name, Database Name, File Paths.

Creating a Parameter

Home Tab -> Manage Parameters -> New.

Name: ServerParameter.

Type: Text.Current

Value: localhost.

Using Parameters

Go to the Source step of your query.

Replace "localhost" with Parameter: ServerParameter.

Connection Strings

Hardcoding is bad. 

- Avoid typing the Server Name in every single query.

Best Practice: Use Parameters 

1. Create a Parameter in Power BI: ServerName, DatabaseName.
2. Use these parameters in your Source step.
3. **Benefit:** Easily switch between DEV (Test DB) and PROD (Live DB).

Credentials & Security

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1. Least Privilege Principle

- The DB User for Power BI should have **READ ONLY** access.
- Never use sa (System Admin) or root credentials!

2. Sharing PBIX files

- Credentials are **NOT** stored in the .pbix file.
- When you send the file to a colleague, they must enter their own credentials to refresh.

Live Demo

Summary

- Views: Our primary way to ingest data.
- Folding: The key to performance. Let the DB work.
- Parameters: The key to flexibility.
- Power Query: The ETL tool where we apply final touches.

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